

MILESTONE 2 · DATASET PREPARATION REPORT

AI-Powered Driver Wellness and Safety Monitoring System

Identifying, cleaning, and structuring model-ready datasets across five driver-monitoring modules ahead of Milestone 3.

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From Drowsiness Detection to Driver Wellness

In Milestone 1, project scope widened from a narrow drowsiness detector into a full driver wellness and safety monitoring system.

Milestone 2 turns that direction into practice: it prepares the datasets each module needs so Milestone 3 can begin model development without delay.

The goal here is not training — it's proof that every dataset is available, understood, cleaned, split without leakage, and structured for direct use next milestone.



Continuous Video

Fatigue detection over time



Facial Landmarks

Temporal head-pose & eye features



Activity Images

Distraction classification



Annotated Frames

YOLO object detection

Milestone 2 Objectives

- ✓ Identify and verify dataset sources
- ✓ Understand size, class distribution & metadata
- ✓ Remove corrupted, duplicate or invalid samples
- ✓ Prevent leakage across splits
- ✓ Document preprocessing for reproducibility
- ✓ Document ownership, format, classes & usage constraints
- ✓ Perform EDA and quality checks
- ✓ Build train / validation / test splits
- ✓ Structure processed folders for direct reuse
- ✓ Confirm readiness for Milestone 3 experiments

Five Modules, Five Owners

Team Member	Module	Dataset / Source	Target Task
Shiwani	Landmark-Based Temporal Features	YawDD	EAR, MAR & head-pose sequence modeling
Kushagra	Video-Based Fatigue Detection	UTA-RLDD	Safe / Caution / High-Risk classification
Shubham	Driver Activity Classification	AUC Distracted Driver	Distraction image classification
Sohini	Seat Belt & Phone Detection	DMS Dataset	YOLO object detection
Ravina	Smoking & Drinking Detection	YOLO smoking/drinking set	Object detection

Dataset Selection at a Glance

SHIWANI

YawDD

Yawning & face videos → EAR, MAR, pitch, yaw, roll. Output: 30-frame windows, 5 features/frame.

KUSHAGRA

UTA-RLDD

Fold1_part1 & part2 — 12 subjects, 36 videos. Labels mapped to Safe / Caution / High-Risk.

SHUBHAM

AUC Distracted Driver

5 activity classes incl. texting, talking on phone, turning, safe driving.

SOHINI

DMS Dataset

Filtered to a clean 2-class YOLO set: Phone, Seatbelt.

RAVINA

Smoking / Drinking Set

YOLO-format images, labels & data.yaml, standard train/val/test layout.

A Common Inspection Process, Every Dataset

Before any preprocessing began, all five datasets passed through the same verification pipeline — so quality issues were caught early and consistently.

1

Verify folder structure & class labels

2

Count samples per class

3

Check for missing or unreadable files

4

Generate class-distribution charts

5

Visualize sample images / frames

6

Review resolution, FPS, duration, brightness

7

Flag duplicates, corrupt files, bad annotations

8

Confirm train / validation / test splits are feasible

RESULTS

Data Quality, By the Numbers

99.35%

landmark extraction success across
349 videos / 288,174 frames (YawDD)

36 / 36

UTA-RLDD videos verified present &
readable — zero corrupt files

4,307

clean activity images retained after
removing 2,969 duplicates + 10
corrupt files

6,701

bounding boxes validated across
6,140 seat-belt / phone images &
labels

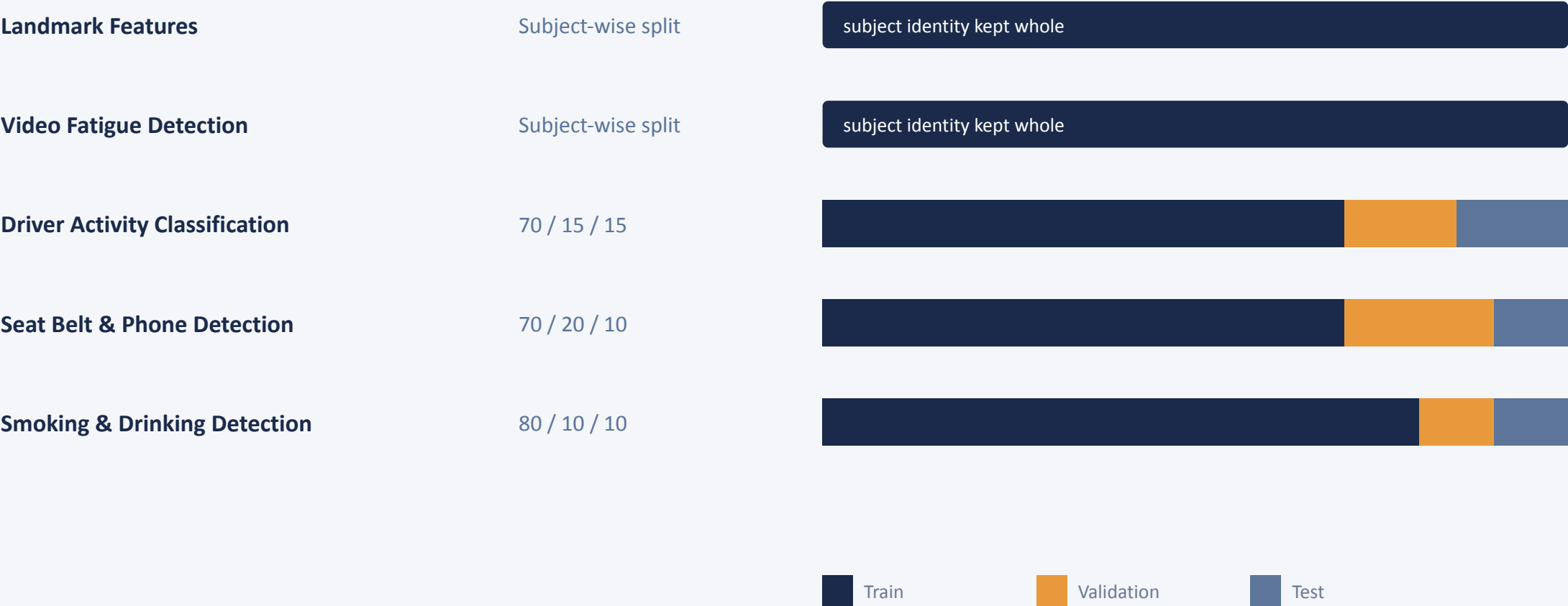
Duplicate detection used perceptual hashing; corruption checks used OpenCV load tests; blur was scored via Laplacian variance — blurry-but-valid frames were kept to preserve realistic driving conditions.

What Happened to Each Dataset

Module	Key Preprocessing Steps
Driver Activity (images)	Resize to 224×224 · normalize · augment with flip, brightness/contrast, rotation
Video Fatigue (UTA-RLDD)	Extract 5 frames/video at 10/30/50/70/90% timeline → 180 processed frames
Landmark Features (YawDD)	Extract EAR, MAR, pitch, yaw, roll · drop invalid frames · 30-frame windows, stride 15
Seat Belt / Phone (YOLO)	Resize to 640×640 · validate boxes · remap classes to Phone / Seatbelt
Smoking / Drinking (YOLO)	Class harmonization · deduplication · augmentation · split verification

Train / Validation / Test Strategy

Split method follows the leakage risk of the data type — subject-wise where identity could leak, ratio-based where samples are independent.



Preventing Leakage Across Splits

Landmark Module

All videos from one subject stay in a single split; 30-frame windows never cross splits; normalization stats come from training data only.

Fatigue Detection

Strict subject-level split (01–08 train, 09–10 val, 11–12 test) — frames extracted only after the split was finalized.

Driver Activity

Duplicates removed pre-split; a post-split hash check flagged near-duplicate frames for future sequence-wise splitting.

Seat Belt & Phone

Original split boundaries preserved — no random frame migration between near-identical consecutive frames.

Smoking & Drinking

Deduplication before splitting; augmentation applied to the training set only, so augmented copies never leak into val/test.

Processed Datasets, Ready to Load

Landmark Features

```
processed_dataset/  
├ raw_features_all_videos.csv  
├ normalized_features_all_videos.csv  
├ train_windows.npy  
├ val_windows.npy  
├ test_windows.npy  
├ train_meta.csv  
├ val_meta.csv  
└ test_meta.csv
```

Video Fatigue Detection

```
data/processed/fatigue_detection/  
├ train/  
│   ├── Safe/  
│   ├── Caution/  
│   └ High_Risk/  
├ val/ (same 3 classes)  
└ test/ (same 3 classes)
```

YOLO Detection Sets

```
<task>_yolo/  
├ data.yaml  
├ train/  
│   ├── images/  
│   └ labels/  
├ valid|val/  
│   ├── images/  
│   └ labels/  
└ test/  
    ├── images/  
    └ labels/
```

Seat-belt/phone and smoking/drinking detection share this layout; activity classification mirrors it with class-named folders instead of images/labels.

Final Processed Dataset Summary

Owner	Feature	Final Format	Split	Leakage Control
Shiwani	Landmark features	.npy + .csv	Subject-wise	No shared windows; train-only normalization
Kushagra	Video fatigue	Class-wise folders	Subject-wise	Subject split before frame extraction
Shubham	Activity classification	Image folders	70/15/15	Duplicates removed before split
Sohini	Seat belt & phone	YOLO + data.yaml	70/20/10	Sequence-based stratified split
Ravina	Smoking & drinking	YOLO + data.yaml	80/10/10	Dedup + train-only augmentation

Model Input Readiness

Feature	Planned Model(s)	Input Shape	Status
Driver Activity	ResNet50, MobileNetV3, EfficientNet-B0	(3, 224, 224) RGB	Ready
Video Fatigue	CNN-LSTM, CNN-GRU, TCN, 3D CNN	(16, 224, 224, 3)	Ready for sequencing
Landmark Features	LSTM	(30, 5)	Ready
Seat Belt / Phone	YOLOv8n, YOLO11n	(640, 640, 3) + labels	Ready
Smoking / Drinking	YOLO-based	(640, 640, 3) + labels	Ready

Reproducibility and Hosted Datasets

REPRODUCIBILITY PLAN

- 1 Download raw dataset from documented source
- 2 Run the dataset inventory script / notebook
- 3 Validate missing files, corrupt files, annotations
- 4 Generate metadata and EDA charts
- 5 Apply cleaning (dedup, invalid-frame removal)
- 6 Create train / validation / test splits
- 7 Save processed files into final folder structure
- 8 Use hosted processed dataset where available

HOSTED DATASET STATUS

Shiwani	Landmark Features	Provided
Kushagra	Video Fatigue Detection	Provided
Shubham	Activity Classification	Provided
Sohini	Seat Belt & Phone	Provided
Ravina	Smoking & Drinking	Provided

Where a hosted link isn't ready yet, the dataset can be regenerated from the documented preprocessing scripts.

Adequacy and Known Limitations

Datasets are adequate for Milestone 3 — every planned model has a matching processed dataset. A few constraints remain to keep in mind during training and evaluation:

- Some datasets reflect controlled, not real-world, environments
- Night, rain, glare and occlusion cases may be underrepresented
- Fatigue detection uses a subset of UTA-RLDD (full set is large)
- Activity data may contain near-duplicate consecutive frames
- Some datasets lack subject IDs, limiting perfect leakage control
- Hosted links still need verification before final submission
- Smoking/drinking stats should expand as more data is analyzed

Roadmap to Milestone 3

1

Train image classifiers for driver distraction

2

Train / fine-tune YOLO for seat-belt, phone, smoking & drinking

3

Train LSTM models on landmark temporal features

4

Sequence fatigue frames; test CNN-LSTM, CNN-GRU, TCN, 3D CNN

5

Compare accuracy, precision, recall, F1, mAP, FPS & false alarms

6

Integrate module outputs into one driver wellness score

Team Contributions & Conclusion

Member	Milestone 2 Contribution
Kushagra	UTA-RLDD dataset, EDA, split strategy, processed frames, report & GitHub organization
Shiwani	Landmark feature dataset, processed sequence files, dataset hosting summary
Shubham	Driver activity preprocessing, model-readiness summary, input format details
Sohini	Seat belt / phone processing, quality checks, leakage prevention, YOLO structure
Ravina	Smoking / drinking dataset structure, hosting details, presentation & final review

Milestone 2 sets the dataset foundation for the AI-Powered Driver Wellness and Safety Monitoring System — five modules, all EDA-checked, cleaned, split without leakage, and structured for direct use in Milestone 3 model training.